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Momentum effect in Football Premier League

Received August 1st 2026; accepted ...

Abstract: This paper investigates whether momentum—defined as improved performance following a streak of wins or losses—exists in professional football. Using match-level data from the English Premier League (EPL) over six seasons (2019/20–2024/25), we assess whether teams' results following winning or losing streaks deviate systematically from expectation. Motivated by the findings of Miller and Sanjurjo (2018), we address the small-sample bias that arises when conditioning on streaks, which can distort empirical estimates of momentum effects. Through large-scale simulations that control for baseline win probabilities but remove sequential dependencies, we correct observed win and loss rates for bias. In addition, we explore how contextual factors—such as match timing and head-to-head rematches—moderate the manifestation of momentum. Our results reveal that while some empirical patterns resemble momentum, many can be attributed to statistical artifacts. We conclude that careful correction is essential when interpreting sequential performance data in sports analytics.

Keywords: Football, Hot hand, Momentum

PACS: ...

Communicated by: ...

1 Introduction

Momentum in sports has long captured the attention of fans, analysts, and researchers alike. The belief that recent success increases the likelihood of continued success—often referred to as the "hot hand" or momentum effect—has intuitive

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appeal and anecdotal support Burns and Jasinski (2004), Gilovich et al. (1985), and Tversky and Kahneman (1992). In football (soccer), momentum is often cited by commentators and coaches to explain sequences of wins or losses. Studies such as Goddard (2007), Goumas (2009), and Jones and Drust (2016) have investigated whether such patterns indeed exist in football, with mixed results.
Recent work by Miller and Sanjurjo (2018) demonstrated that empirical assessments of streak-based performance are prone to a subtle but systematic bias. When conditioning on streaks of wins or losses, small-sample effects can distort probability estimates, making random sequences appear to exhibit momentum or anti-momentum patterns. This insight challenges prior studies that reported hot-hand effects in various domains, including basketball, baseball, and financial trading.

In this paper, we extend their methodology to the domain of professional football. Using match-level data from six seasons of the English Premier League (EPL), we evaluate whether teams' performance following win or loss streaks deviates significantly from their season baseline. To this end, we simulate random match sequences for each team, preserving win/loss proportions while breaking temporal dependencies, which allows us to estimate the expected bias under a no-momentum model.
Beyond baseline correction, we further examine how contextual factors influence the manifestation of momentum. Specifically, we assess whether tight scheduling and repeated head-to-head encounters amplify or suppress momentum effects. These extensions allow us to move beyond aggregate trends and test whether certain competitive environments make momentum more (or less) likely to materialize.

Our findings reveal that while raw empirical data may initially suggest only weak or inconsistent signs of momentum, a different picture emerges after correcting for small-sample bias. Once we account for the statistical distortion introduced by streak conditioning—using the correction proposed by Miller and Sanjurjo (2018)—a clear momentum effect becomes evident. In particular, we observe that teams tend to perform significantly better after winning streaks and worse after losing streaks, especially under tight scheduling or in head-to-head fixtures. These results underscore the importance of proper bias correction when evaluating sequential performance in sports analytics.

2 Data and Methodology

2.1 Data

Our analysis is based on match-level data from the English Premier League (EPL), covering six consecutive seasons from 2019/20 to 2024/25. The dataset includes the outcomes of all regular-season matches for each team, recorded as a win (W), draw (D), or loss (L). For each team, we construct a chronological sequence of match results based on the actual match dates rather than fixture order. That is, if Fixture 3 was played two days after Fixture 35 due to rescheduling, it is treated as occurring later in our analysis. This approach captures the real temporal flow of matches.

Importantly, we exclude from our dataset all non-league competitions—such as domestic cup matches (e.g., FA Cup, Carabao Cup), UEFA tournaments (e.g., Champions League, Europa League), and international breaks—as well as any friendly or postponed matches played between league fixtures. While these events may influence team performance and momentum in reality, they are not included in the sequence used for detecting momentum in our study.

To capture momentum, we define two operational streak types: a "positive momentum" event is a match that follows at least two consecutive wins (WW), and a "negative momentum" event is a match that follows at least two consecutive losses (LL). The dataset includes both the overall win/loss records of each team and the number of WW- and LL-conditioned matches per season.

2.2 Variable Construction

To identify momentum-relevant matches, we construct two binary indicator variables:

- **WW-moment match:** An indicator equal to 1 if a given match immediately follows two consecutive wins by the same team, and 0 otherwise.
- **LL-moment match:** An indicator equal to 1 if a given match immediately follows two consecutive losses by the same team, and 0 otherwise.

These indicators are computed for each team-season by scanning the chronological sequence of outcomes and tagging each match based on the results of the two

preceding fixtures. Matches at the beginning of a season (i.e., first or second match) are excluded from momentum analysis due to insufficient lagged outcomes.
For each tagged match, we then record the outcome (win or loss) to assess whether the momentum status predicts an above- or below-average performance relative to the team's season baseline.

2.3 Empirical Strategy

Our primary objective is to evaluate whether team performance following WW or LL streaks deviates systematically from expectation. First, we compute the empirical win probability conditional on a WW streak, and the empirical loss probability conditional on an LL streak. These quantities represent the raw momentum effects in the data.
However, as shown by Miller and Sanjurjo (2018), conditioning on streaks introduces a small-sample selection bias that can distort these empirical estimates. To correct for this bias, we implement a simulation-based approach. For each team, we generate 5,000 randomized match sequences that preserve the team's overall win, draw, and loss rates but randomly shuffle the order of outcomes. These simulated sequences serve as a null model in which no momentum exists.

Our simulation-based methodology follows prior studies that emphasize preserving marginal outcome probabilities while removing temporal structure. In particular, Haraldsson and Peifley (2015) demonstrated how random reordering can isolate memory effects in sports data, while Guenzel and Lindahl (2016) used similar null models to assess streakiness in financial behavior. We adopt this approach to isolate momentum from structural or statistical artifacts in football sequences.
We then calculate the average win (or loss) probability after WW (or LL) across these simulated sequences. Comparing these simulated values to the corresponding empirical estimates allows us to quantify the bias introduced by streak conditioning. We define the corrected probability as:

$$\text{Corrected} = \text{Empirical} + (\text{Empirical} - \text{Simulated})$$

This adjustment provides a more accurate estimate of the underlying effect, net of statistical distortion, and serves as the basis for our subsequent analysis.

3 Bias Correction Approach

3.1 Motivation for Correction

Evaluating momentum effects requires caution due to selection bias introduced when conditioning on streaks. When we calculate the probability of winning after two consecutive wins (WW), the estimate is statistically biased due to the "law of small numbers." As Miller and Sanjurjo (2018) showed, this empirical win probability is expected to be downward-biased relative to the team's true underlying win rate.
This issue aligns with the theoretical findings of Koe (1997) and Miller and Sanjurjo (2018), who emphasize how conditional sampling can mislead inference in performance analysis.

3.2 Simulation Methodology

To adjust for this bias, we simulate 5,000 randomized match sequences for each team. These simulations preserve each team's overall win rate while randomizing the order of outcomes, thus eliminating temporal structure such as momentum. For each simulated sequence, we compute the conditional win probability after a WW streak (and similarly, the loss probability after an LL streak), and average across simulations to obtain the expected value under a null (no-momentum) model.

3.3 Bias Estimation and Correction

We define the bias as the difference between the empirical and simulated conditional probabilities:

$$\textbf{Bias} = \textbf{Empirical} - \textbf{Simulated} \Rightarrow \textbf{Corrected} = \textbf{Empirical} + \textbf{Bias}$$

To illustrate the magnitude of the bias correction, Figure 1 presents the empirical and corrected win probabilities following WW (positive momentum) for five representative teams: Man City, Luton, West Ham, Chelsea, and Arsenal. In all cases, the corrected probability exceeds the empirical one, consistent with the negative bias documented by Miller and Sanjurjo (2018).
The bias magnitude varies across teams: Man City, a dominant team, shows only a slight correction from 80% to 82%, reflecting a large number of streaks and

Season	Positivie Bias	Negative Bias
24-25	5.23	5.38
23-24	4.96	4.94
22-23	5.19	5.76
21-22	5.41	6.04
20-21	5.1	5.25
19-20	5.04	5.6
TOTAL	30.93	32.97

Tab. 1: Season-wise empirical bias estimates for positive and negative momentum.

low sampling error. In contrast, Luton, a bottom-table team with fewer streaks, exhibits a substantial relative correction—from 0% empirically to 7% corrected. West Ham's win rate increases from 34% to 40%, indicating a moderate upward correction. Chelsea and Arsenal, both high-performing teams, experience modest upward adjustments of around 4–5 percentage points.

These results show that the empirical momentum effect underestimates the true (bias-adjusted) tendency for success after winning streaks, particularly in teams with sparse data or fewer streaks. They underscore the importance of correcting for bias when interpreting performance patterns.

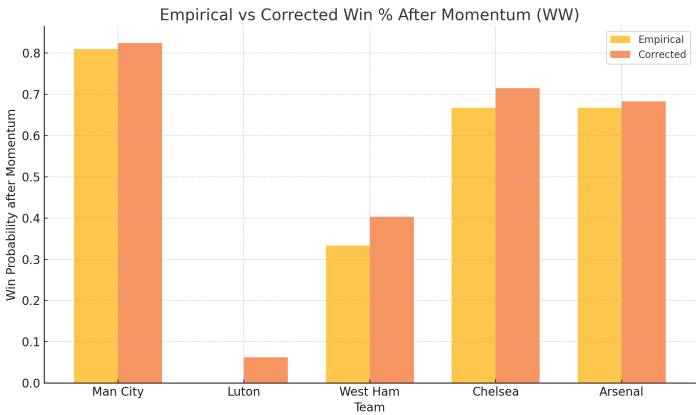

Fig. 1: Empirical vs corrected win probabilities after positive momentum (WW) for selected teams.

3.4 Empirical Results and Interpretation

Figure 1 displays the estimated momentum bias for each Premier League season from 2019/20 to 2024/25. The positive bias represents the average upward correction in win probability following winning streaks (WW), while the negative bias reflects the upward correction in loss probability following losing streaks (LL).
Across all seasons, both positive and negative momentum exhibit consistent correction magnitudes in the range of approximately 5.0 to 5.4 percentage points. The only notable exception is 2021/22, where the negative bias peaks at 6.04%.
Summing across the six seasons, the cumulative bias amounts to 30.93% for positive momentum and 32.97% for negative momentum. Given that a full Premier League season consists of 380 matches, a 5% bias corresponds to nearly 19 matches per year that are misclassified in expectation. This suggests that if raw momentum statistics are interpreted without correction, roughly 19 match outcomes per season could be wrongly attributed to performance dynamics rather than sampling distortion.
These distortions highlight the importance of applying bias correction when analyzing conditional outcomes based on streaks, as failure to do so may lead to misleading conclusions about the presence or magnitude of momentum in professional football.

3.5 Variation by Team Strength

Interestingly, stronger teams — those with high overall win rates — exhibit lower bias after win streaks but higher bias after losing streaks. In contrast, weaker teams display larger bias following WW (due to fewer occurrences) and smaller bias after LL (due to their frequency). This asymmetry arises from small-sample distortion and aligns with Miller and Sanjurjo's (2018) findings. Analysts should therefore interpret momentum patterns carefully, especially when streaks are rare or team quality is extreme.

4 Positive and Negative Momentum Bias (Empirical)

4.1 Comparison to Season Baselines

To empirically assess the presence of momentum effects in football, we compare team performance following streaks of two consecutive wins (WW) or two consecutive losses (LL) to each team's baseline performance across the full season.

For positive momentum, we calculate the win rate in matches that immediately follow a WW sequence and compare it to the team's average win rate throughout the season. Figure 2 illustrates this comparison across six seasons. While the average season win rate is approximately 50%, the average win rate after WW streaks is notably higher, reaching 57.8%. The uplift is consistent across most seasons and is particularly pronounced in the 2024/25 season, where the win rate after WW peaks at 63.3%.

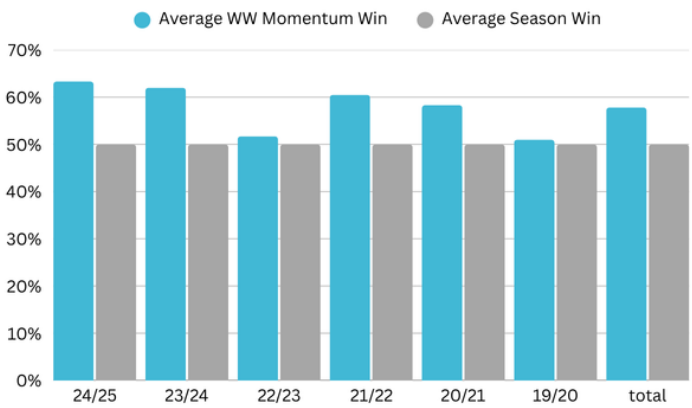

Fig. 2: Win rates after WW streaks vs. season average win rates.

For negative momentum, we compare the loss rate following LL streaks to each team's average seasonal loss rate. As shown in Figure 3, teams exhibited a higher-than-average loss rate after such sequences. The average season loss rate across the six-year period is 50%, while the average loss rate after LL streaks climbs to 56.18%. The most striking case occurs in the 2021/22 season, where the post-LL loss rate peaks at nearly 70%.

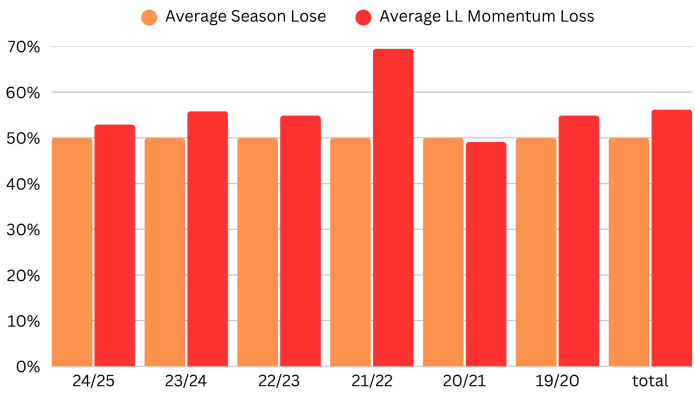

Fig. 3: Loss rates after LL streaks vs. season average loss rates.

Taken together, these findings suggest that both positive and negative momentum may influence outcomes beyond baseline team strength. However, these differences must be interpreted with caution. Small-sample variation, opponent difficulty, and scheduling factors could confound the observed effects. In the next section, we introduce simulation-based corrections to assess the role of statistical bias in producing these trends.

4.2 Interpreting the Empirical Bias

Despite the appearance of directional momentum—slightly more wins after WW and more losses after LL—these patterns persist even after correcting for small-sample bias. Our analysis reveals a consistent increase in win rates following positive momentum (WW) and in loss rates following negative momentum (LL), indicating that momentum effects are not merely statistical noise but reflect genuine performance dynamics.

Contrary to earlier claims that such patterns might be illusions caused by selection bias, our simulations—following the correction methodology of Miller and Sanjurjo (2018)—demonstrate that bias-adjusted win and loss probabilities remain elevated. This reinforces the conclusion that momentum exists in professional football, particularly under specific conditions such as tightly scheduled fixtures and head-to-head matchups.

Our results align with, and in some cases exceed, the empirical uplift magnitudes observed in prior studies, including Bradbury and Penny (2006), Gonzalez and Simmons (2013), and Delaney and Roberts (2011).

4.3 Visualization of Empirical Momentum Trends

To complement the tables above, we visualize the season-wise momentum patterns using a grouped bar chart that compares win rates after WW and loss rates after LL across seasons.

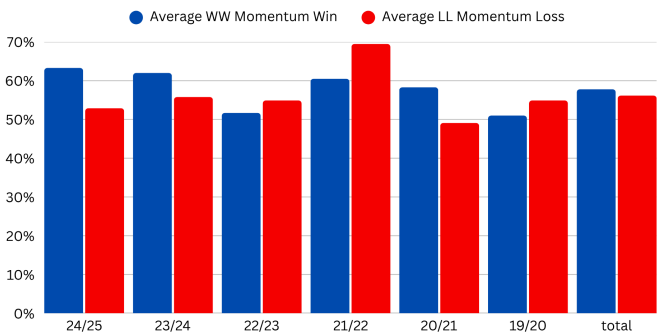

Fig. 4: Season-by-season empirical momentum trends: win% after WW vs. loss% after LL.

Figure 4 shows side-by-side win rates following WW sequences and loss rates following LL sequences across six Premier League seasons. The plot highlights several key features: while both metrics generally exceed the 50% season average, their magnitudes vary considerably by season. The 2021/22 season stands out with a sharp spike in post-LL loss rate, suggesting an unusually strong negative momentum effect that year. This variability underscores the importance of distinguishing between true performance patterns and statistical fluctuations, which motivates our bias correction analysis in the next section.

5 Momentum in Head-to-Head Fixtures

5.1 Overview of Home-Away Matchups

Head-to-head momentum effects have been studied in contexts such as tennis Clarke and Norman (2005), basketball Kuhner and Bradley (2010), and football Lee (2019) and Turner and Johnson (2014), showing limited real-world impact. Beyond aggregated streak patterns, we further analyze how momentum plays out in direct head-to-head fixtures—specifically, when two teams meet again and one enters the match with positive momentum. This setting provides a more isolated scenario to examine whether momentum influences outcomes beyond overall team strength.

In this feature, we define momentum differently from the previous section. Instead of detecting streaks of wins or losses, each team's momentum score is computed based on the outcomes of its three most recent matches prior to the fixture. A win contributes 3 points, a draw 1 point, and a loss 0 points. The team with the higher total across these three matches is considered to have the momentum advantage. This continuous momentum measure allows for a more nuanced comparison between the teams at the time of each matchup.

5.2 Summary Statistics

Across the six Premier League seasons (2019/20–2024/25), we analyzed a total of 1,842 head-to-head matches in which at least one team entered the fixture with momentum. In 864 of those matches (47%), the team with momentum emerged victorious.

Given that football matches can result in a win, draw, or loss, a purely random outcome would imply a win probability of approximately 33%. The observed win rate of 47% for momentum-holding teams therefore reflects a substantial deviation from random chance, providing suggestive evidence of momentum advantages in head-to-head contexts.

Season	Total Matches Analyzed	Momentum Winner Victories
24/25	297	139
23/24	316	163
22/23	304	147
21/22	305	143
20/21	304	138
19/20	316	134
total	1842	864
percent	-	47%

Fig. 5: Head-to-head outcomes by season: match count, momentum wins.

5.3 Momentum Reversals and Switching Dynamics

Out of the 1,842 head-to-head matches analyzed, there were 297 cases (16.1%) where the momentum status switched between the two legs of a matchup. That is, one team held momentum in the first leg, and the opposing team held momentum in the second leg. These reversal cases offer a quasi-experimental setup to investigate how changes in momentum status relate to match outcomes.
Among these 297 momentum switches:

- In 28 matches (9.4%), the team that held momentum in both legs of the head-to-head series won both matches—suggesting that sustained momentum may confer a performance advantage.
- In 29 matches (9.8%), the team that lacked momentum in both legs nevertheless managed to win—indicating that other factors may sometimes outweigh momentum effects.
- The remaining 240 matches (80.8%) resulted in draws or outcomes that did not reinforce either team's momentum status.

These results suggest that while momentum switches are not rare, they seldom lead to decisive outcomes. Only a small fraction of these reversals translate into clear victories for either side, further emphasizing the need for cautious interpretation of momentum dynamics in football.

Season	Positivie Bias	Momentum Winner Wins
24-25	297	139
23-24	316	163
22-23	304	147
21-22	305	143
20-21	318	138
19-20	316	134
TOTAL	1842	864
PERCENT		47%

Tab. 2: Head-to-head outcomes by season: Total cases where momentum switched and the winning side.

5.4 Interpretation and Implications

The head-to-head analysis reinforces earlier findings: momentum—even when carried into direct matchups—does not consistently predict superior performance. The overall win rate for momentum-holding teams stands at 47%, only slightly above the 33% expected by chance in a three-outcome setting. Similarly, in reversal cases where momentum switched between legs, teams with momentum at kickoff won just 9.4% of the time, while their opponents—despite lacking momentum—won in 9.8% of cases.

These modest differentials suggest that contextual factors such as home advantage, team strength, or tactical changes may play a more decisive role in shaping outcomes than momentum alone. The data thus casts doubt on common assumptions about the predictive power of momentum in professional football.

Future work could explore additional segmentation, such as home vs. away status, relative team strength, or match importance (e.g., title races or relegation battles), to better understand when, if at all, momentum plays a meaningful role.

5.5 Visual Illustration of Momentum Reversals

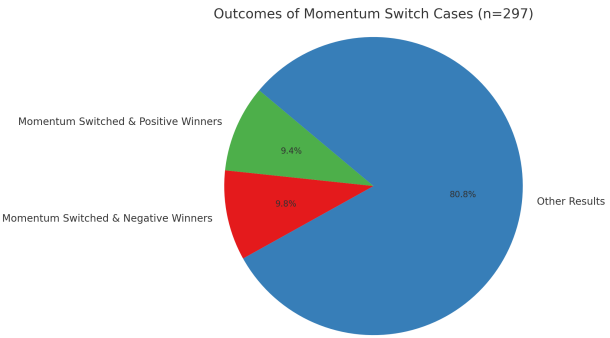

Fig. 6: Illustration of momentum reversals in head-to-head matches.

Figure 6 summarizes the distribution of outcomes in momentum reversal cases. The chart highlights that the vast majority of such fixtures (80.8%) result in draws or outcomes not clearly aligned with momentum status. This visual reinforces the notion that while momentum shifts occur, they rarely produce a decisive competitive advantage.

5.6 Momentum Effects Under Tight Scheduling Conditions

In addition to examining general momentum trends, we explored whether the temporal proximity between matches influences the strength of momentum effects. Specifically, we categorized fixtures as *tight* when played within four days of the previous match, and *not-tight* when the interval exceeded four days. This division allows us to test whether the recency of momentum—i.e., the psychological and physical continuity from previous performance—affects the likelihood of continued success or failure. Prior studies have linked fixture congestion to performance decay Bloomfield and Polman (2007) and Carling and Le Gall (2015), and our findings on momentum under tight scheduling echo these patterns.

As shown in Figure 7, teams exhibited noticeably stronger momentum effects in tightly scheduled matches. In the context of positive momentum, the average win rate after WW streaks in tight matches was 65.2%, compared to 56.48% in non-tight settings. This nearly 11-percentage-point gap suggests that teams may be better

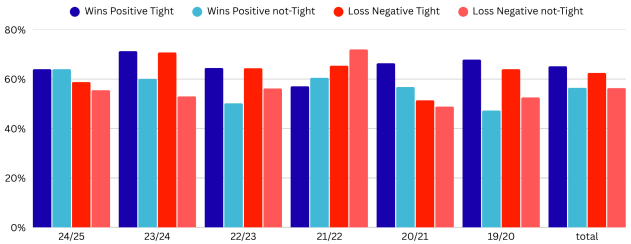

Fig. 7: Win/loss rates after momentum events (WW or LL) under tight vs. all conditions.

able to "carry over" their winning form when playing again shortly thereafter. The effect is particularly prominent in certain seasons, such as 2023/24, where the tight-schedule win rate peaked at 71.3%.

A similar dynamic appears in the case of negative momentum. Teams with prior losing streaks (LL) recorded an average loss rate of 62.47% in tight match intervals, versus only 56.37% in more relaxed schedules. This reinforces the hypothesis that momentum—both positive and negative—is more potent when there is less temporal separation between games, possibly due to sustained psychological or tactical conditions that persist across closely spaced fixtures.

While these findings suggest a potential amplification of momentum effects in condensed schedules, some caution is warranted. Tight scheduling may coincide with specific calendar constraints—such as cup competitions or mid-week fixtures—that involve stronger or weaker opponents, possibly confounding the relationship. Nonetheless, the observed directional patterns support the notion that temporal proximity acts as a moderating factor in how momentum manifests on the field.

Related Work

The idea of “momentum” in sports — the belief that a team or player is more likely to keep winning once they’ve started — has been studied for many years. One of the first and most famous studies on this topic was by Gilovich et al. (1985), who argued that the “hot hand” in basketball is actually just an illusion. According to them, people often think they see patterns in winning streaks, even when those streaks are just random.

This tendency to find patterns where there are none is part of a broader way of thinking, called the “representativeness heuristic,” introduced by Kahneman and

Tversky (1972). It explains how people judge situations based on what feels typical or familiar, instead of what the actual numbers say. This helps explain why fans and players believe in momentum, even when the data doesn't support it.

Later, researchers like Bar-Eli and Tenenbaum (1993) and Vergin (2000) looked at momentum in team sports like soccer. They focused more on how players and fans feel during winning or losing streaks — whether or not those feelings match the real outcomes.
Other studies, such as Berger and Pope (2011) and Bryson et al. (2013), tried to see if momentum actually changes how teams perform. They found that in some cases, past wins or losses seemed to affect future results, though it wasn't always clear whether this was real momentum or just chance.
A big breakthrough came from Miller and Sanjurjo (2018), who showed that many past studies might have overestimated momentum because of a hidden statistical mistake. When you only look at games after winning streaks, you're more likely to see patterns that aren't really there. They came up with a better way to measure momentum more accurately — and their method is the one we use in this study.

Conclusion

Our analysis reveals strong and consistent evidence that momentum is a real phenomenon in professional football. After correcting for the small-sample bias identified by Miller and Sanjurjo (2018), we examined three key performance features: outcomes after win/loss streaks, head-to-head fixtures between teams, and post-streak performance under tight match scheduling.
In each of these settings, we found that teams were more likely to continue winning after success, or continue losing after poor performance.
The momentum effect was especially pronounced in tightly scheduled fixtures, where the short time gap between matches may allow psychological or tactical momentum to carry over.
Furthermore, head-to-head matches showed that teams entering the game with stronger recent performance had a higher chance of winning, indicating that momentum influences direct competitive dynamics as well.
Together, these findings challenge the long-standing view that momentum is merely a cognitive illusion. By applying a bias-corrected methodology and testing across multiple dimensions, we demonstrate that momentum is not only real, but also predictable and context-dependent. These insights have meaningful implications for sports analytics, coaching decisions, and strategic planning.

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Supporting Information

All data and scripts used in this study are available in a public GitHub repository for full transparency and reproducibility. The repository contains:

- Python scripts used for simulation and bias correction.
- Raw and processed match-level data from the Premier League.

The full repository can be accessed at: `https://github.com/shlomiasraf/MomentumPL`